Presented by : Analytics in Practice Group 31

BUSINESS CREDIT APPLICATION PREDICTON

Improving Credit Decision Efficiency and Customer Experience

01

WHO WE ARE
&
WHY WE ARE HERE

- We specialise in predictive modelling and data-driven decisions.
- Faster decision making for managers with lower risks and operation costs.
- Our aim : improve retention of high-potential customer segments

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BUSINESS GOAL

PREDICT FUTURE CREDIT APPLICATIONS

- Understanding Emerging Customer Credit Needs
- Data-Driven Credit Demand Forecasting

01 **Identify Likely Applicants**

02 **Proactive Customer Retention**

03 **Improved Relationship Management**

CURRENT CHALLENGES

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CLASS IMBALANCE

- Small proportion of credit applicants (9 : 1 ratio)

LIMITED VISIBILITY ON CUSTOMER INTENT

- No clear signals on upcoming business credit needs

REACTIVE ENGAGEMENT

- The bank only reaches customers after they apply for credit

LOSING VALUABLE CUSTOMERS

- High-value customers may switch to competitors for better credit options

WHAT CAN OUR MODEL DO

1. Understanding Customer Behaviour Patterns
2. Predict Likelihood of Future Credit Needs
3. Enable Proactive Customer Engagement
4. Support Customer Retention Efforts
5. Prioritise Relationship Manager Efforts
6. Improve Targeting for Business Credit Campaigns

OUR APPROACH

Step 1

System reads customer business activity

Step 2

System analyses the historical pattern

Step 3

System predicts likelihood for future applications

Step 4

Managers receive a prioritise list of customers

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MODELS COMPARED

We tested several approaches to ensure the solution is reliable under class imbalance and robust for real business use.

01 Logistic Regression

02 Random Forest

03 SVM

04 XGBoost

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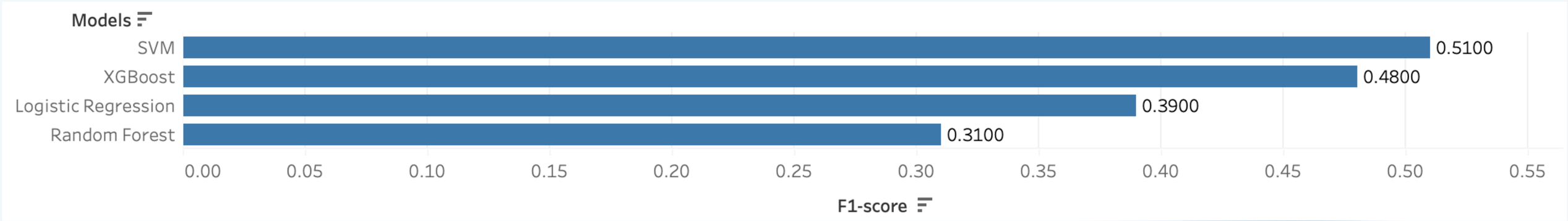
MODEL RESULTS



F1 score is a machine learning evaluation metric that measures a model's accuracy

KEY BENEFITS FROM US

- Gain a predictive edge by identifying credit-ready customers earlier than competitors.
- Strengthen customer loyalty through timely, proactive engagement.
- Allocate resources more efficiently for faster, smarter decisions.
- Capture high-value opportunities before competitors even recognise them.

PREDICTED IMPROVEMENTS

- Identify about 2 out of 3 customers who are likely to apply for credit early.
- Capture 4–6 times more real credit opportunities than current methods.
- Reduce missed high-value customers by over 60%.
- Focus manager time on customer who are more likely to take credit offers
- Make faster, confident decisions with highly reliable predictions (≈90% consistency)
- Improve team efficiecny and reduce wasted efforts by 40%

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THANK YOU